

tu 1

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1 Smooth Functions

1.7. Write $f(x, y) = x^2g_{11} + xyg_{12} + y^2g_{22}$, so for $t \neq 0$, we have

$$\begin{aligned} g(t, u) &= \frac{1}{t}(t^2g_{11}(t, tu) + (t^2u)g_{12}(t, tu) + t^2u^2g_{22}(t, tu)) \\ &= tg_{11}(t, tu) + tug_{12}(t, tu) + tu^2g_{22}(t, tu), \end{aligned}$$

and for $t = 0$, $g(t, u)$ equals the same expression. Thus, $g(t, u) \in C^\infty$.

Claim: There does not exist an embedding $f : \mathbb{R}P^n \rightarrow \mathbb{R}^n$ for $n \geq 1$.

Proof: Since $\mathbb{R}P^n$ is compact, and $\mathbb{R}^n$ is Hausdorff, f is a closed map. It is enough to require f injective. This condition also gives $\text{im } f$ is compact and therefore closed. I claim that $\text{im } f$ is also open. Pick $x \in \mathbb{R}P^n$ and $U(x) \cong \mathbb{R}^n$. We have the (injective) restriction $f : U \rightarrow \mathbb{R}^n$, so by Invariance of domain $f(U)$ is open. Then $\text{im } f = \bigcup_{x \in \mathbb{R}P^n} f(U) \neq \emptyset$ is open in $\mathbb{R}^n$, so $\text{im } f = \mathbb{R}^n$ by connectedness of $\mathbb{R}^n$. However, $\mathbb{R}^n$ is not compact.

Claim: If D is a point-derivation of C_p^∞ and f is constant, then $D(f) = 0$.

Proof: $D(1) = D(1)1 + 1D(1) = 2D(1)$, so $D(1) = 0$. Then $D(c) = cD(1) = 0$.

Claim: Let A be an algebra over the field K and D_1, D_2 be derivations of A . Then $D_1 \circ D_2 - D_2 \circ D_1$ is a derivation of A .

Proof: $D_1 \circ D_2 - D_2 \circ D_1$ is linear. Further,

$$\begin{aligned} (D_1 \circ D_2 - D_2 \circ D_1)(ab) &= D_1(D_2(a)b + aD_2(b)) - D_2(D_1(a)b + aD_1(b)) \\ &= D_1D_2(a)b + D_2(a)D_1(b) + D_1(a)D_2(b) + aD_1D_2(b) - D_2D_1(a)b - D_1(a)D_2(b) - D_2(a)D_1(b) - aD_2D_1(b) \\ &= D_1D_2(a)b + aD_1D_2(b) - D_2D_1(a)b - aD_2D_1(b) \\ &= (D_1 \circ D_2 - D_2 \circ D_1)(a)b + a(D_1 \circ D_2 - D_2 \circ D_1)(b) \end{aligned}$$

$$\omega(X)_p = \omega_p(X_p) = \sum_{ij} a_i(p)b^j(p)(dx^i)_p \left(\frac{\partial}{\partial x^j} \right) \Big|_p = \sum_i a_i(p)b^i(p)$$

Claim: If $f \in C^\infty(U)$, then $\omega(fX) = f\omega(X)$.

Proof:

$$\begin{aligned}\omega(fX)_p &= \omega_p \left(\sum_i f(p) b^i(p) \frac{\partial}{\partial x^i} \Big|_p \right) = \sum_i a_i(p) f(p) b^i(p) \\ &= f(p) \sum_i a_i(p) b^i(p) = f(p) \omega(X)_p\end{aligned}$$

2 Manifolds

Claim: Let $\{(U_\alpha, \phi_\alpha)\}$ be an atlas of the locally Euclidean space M . If (V, ψ) is compatible with $\{(U_\alpha, \phi_\alpha)\}$ and (W, σ) is compatible with $\{(U_\alpha, \phi_\alpha)\}$, then $(V, \psi), (W, \sigma)$ are compatible with each other.

Proof: By definition, we have the smooth composition with smooth inverse:

$$\psi(V \cap U_\alpha) \xrightarrow{\psi^{-1}} V \cap U_\alpha \xrightarrow{\phi_\alpha} \phi_\alpha(V \cap U_\alpha),$$

which restricts to the composition:

$$\psi(V \cap W \cap U_\alpha) \xrightarrow{\psi^{-1}} V \cap W \cap U_\alpha \xrightarrow{\phi_\alpha} \phi_\alpha(V \cap W \cap U_\alpha).$$

Similarly, we have the composition:

$$\sigma(V \cap W \cap U_\alpha) \xrightarrow{\sigma^{-1}} V \cap W \cap U_\alpha \xrightarrow{\phi_\alpha} \phi_\alpha(V \cap W \cap U_\alpha),$$

so inverting the second composition and post-composing with the first, we have:

$$\psi(V \cap W \cap U_\alpha) \xrightarrow{\phi_\alpha \psi^{-1}} \phi_\alpha(V \cap W \cap U_\alpha) \xrightarrow{\sigma \phi_\alpha^{-1}} \sigma(V \cap W \cap U_\alpha)$$

which is smooth as the composition of two smooth maps. Since the $\{U_\alpha\}$ cover M , we have a smooth map $\sigma\psi^{-1} : \psi(V \cap W) \rightarrow \sigma(V \cap W)$.

Claim: An atlas $\mathcal{U} = \{(U_\alpha, \phi_\alpha)\}$ is contained in a unique maximal atlas $\mathcal{M}$.

Proof: Let $\mathcal{M} = \{(V, \psi) : (V, \psi) \text{ is compatible with } \mathcal{U}\}$. I claim that $\mathcal{M}$ is an atlas, and is the unique maximal atlas containing $\mathcal{U}$. Suppose $(W_1, \sigma_1), (W_2, \sigma_2) \in \mathcal{M}$. By the previous claim, $(W_1, \sigma_1), (W_2, \sigma_2)$ are compatible. Any $(U_\alpha, \phi_\alpha) \in \mathcal{U}$ is compatible with $\mathcal{U}$ and $\bigcup_\alpha U_\alpha = M$. Thus, $\mathcal{M}$ is an atlas containing $\mathcal{U}$. Suppose $\mathcal{M}'$ is an atlas containing $\mathcal{U}$ and $(W, \sigma) \in \mathcal{M}'$. Then (W, σ) is compatible with $\mathcal{U}$, so $(W, \sigma) \in \mathcal{M}$. Thus, $\mathcal{M}$ is maximal.

Claim: The product of two manifolds can be given manifold structure.

Proof: Let M, N be manifolds and $\{(U_\alpha, \phi_\alpha)\}, \{(V_\beta, \psi_\beta)\}$ be their atlases. Let $W_{\alpha\beta} = U_\alpha \times V_\beta$ and $\sigma_{\alpha\beta} = \phi_\alpha \times \psi_\beta$. I claim that $\mathcal{U} = \{(W_{\alpha\beta}, \sigma_{\alpha\beta})\}$ is an atlas of $M \times N$. Indeed, $\bigcup_{\alpha, \beta} (U_\alpha \times V_\beta) = M \times N$,

$W_{\alpha_1\beta_1} \cap W_{\alpha_2\beta_2} = (U_{\alpha_1} \cap U_{\alpha_2}) \times (V_{\beta_1} \cap V_{\beta_2})$, $(\phi_\alpha \times \psi_\beta)^{-1} = \phi_\alpha^{-1} \times \psi_\beta^{-1}$, and compositions of products of functions are done component-wise. The product (in the sense of $\phi \times \psi$) of two smooth functions is smooth, so $M \times N$ is a manifold.

Problem: The sphere with a hair is not a manifold.

Proof: Pick $U(p) \cong \mathbb{R}^n(0)$. If $n > 1$, then $U - p = \mathbb{R}^n - 0$ is connected. However, in the picture $U - p$ is disconnected. $U \neq \{p\}$, so $n \neq 0$, so $n = 1$. We have a homeomorphism h between $V \sqcup W \rightarrow (0, 1) \sqcup (1, 2)$, where $V \leq S^2 - p = \mathbb{R}^2$, $W \leq \mathbb{R}$ are open. WLOG h restricts to $V \rightarrow (0, 1)$. By Invariance of domain, this is absurd.

Problem (Charts on a sphere): Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$. Let $U_1, U_2, \dots, U_6$ be defined as usual. $\phi_1(x, y, z) = (y, z)$, $\phi_2(x, y, z) = (y, z)$, etc.

Ans: We have $U_{14} = \{(x, y, z) \in S^2 : x > 0 \text{ and } y < 0\}$, so $\phi_4(U_{14}) = \{(x, z) \in (D^2)^\circ : x > 0\}$, ie the portion of the open unit disc with positive x -coordinate. We compute $\phi_4^{-1}(x, z) = (x, -\sqrt{1 - x^2 - z^2}, z)$. Thus,

$$\phi_1 \circ \phi_4^{-1}(x, z) = \phi_1(x, -\sqrt{1 - x^2 - z^2}, z) = (-\sqrt{1 - x^2 - z^2}, z),$$

which is smooth in the aforementioned domain.

3 Smooth Maps on Manifolds

Claim (well-definedness of smoothness of a function $f : M \rightarrow \mathbb{R}$ at a point).

Proof: Suppose there exists (U, ϕ) in the atlas with $p \in U$ such that $f \circ \phi^{-1}$ is smooth at $\phi(p)$. Let (V, ψ) be another chart in the atlas. Then $f \circ \psi^{-1} = f \circ \phi^{-1} \circ \phi \circ \psi^{-1}$ is such that $\phi \circ \psi^{-1}$ is smooth and $f \circ \phi^{-1}$ is smooth at $\phi(p)$. Thus, the composition is smooth at $\psi(p)$.

Claim (smoothness of a real-valued function): The following are equivalent. (i). The function $f : M \rightarrow \mathbb{R}$ is smooth. (ii). The manifold M has an atlas such that for every chart (U, ϕ) in the atlas, $f \circ \phi^{-1} : \mathbb{R}^n \supseteq \phi(U) \rightarrow \mathbb{R}$ is C^∞ . (iii). For every chart (V, ψ) on M , the function $f \circ \psi^{-1} : \mathbb{R}^n \supseteq \psi(V) \rightarrow \mathbb{R}$ is C^∞ .

Proof: It is enough to show $(i) \rightarrow (ii) \rightarrow (iii)$. Let $\mathcal{M}$ be a maximal atlas of M , $(U, \phi) \in \mathcal{M}$, and $p \in U$. Pick $(V, \psi) \in \mathcal{M}$ such that $p \in V$ and $f \circ \psi^{-1} : \psi(V) \rightarrow \mathbb{R}$ is C^∞ at $\psi(p)$. Since $f \circ \phi^{-1} = f \circ \psi^{-1} \circ \psi \circ \phi^{-1}$ on $\phi(U \cap V)$ and $\psi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V)$ is C^∞ , we have that $f \circ \phi^{-1}$ is C^∞ at $\phi(p)$. This holds for all $p \in U$, so $f \circ \phi^{-1} : \phi(U) \rightarrow \mathbb{R}$ is C^∞ . If (V, ψ) is a chart, then it is assumed that (V, ψ) is compatible with $\mathcal{M}$, ie $(V, \psi) \in \mathcal{M}$, so the previous result applies????

Claim: If $F : N \rightarrow M, G : M \rightarrow P$ are C^∞ maps of manifolds, so is $G \circ F : N \rightarrow P$.

Proof: Let $p \in N$. Pick charts (X, φ) , (W, ψ) in P, M such that $GFp \in X$ and $Fp \in W$. Thus,

$\varphi G \psi^{-1} : \psi(W \cap G^{-1}X) \rightarrow \varphi(X)$ is C^∞ . Let $Y = W \cap G^{-1}X$. The restriction $(Y, \psi|_Y)$ is a chart in M . Pick a chart (U, ϕ) in N such that $p \in U$. Then $\psi F \phi^{-1} : \phi(U \cap F^{-1}Y) \rightarrow \psi(Y)$ is C^∞ . Thus, the composition $\varphi G F \phi^{-1} : \phi(U \cap F^{-1}Y) \rightarrow \varphi(X)$ is C^∞ .

Claim: Let $U \leq M$ be an open subset of a manifold of dimension n . If $F : U \rightarrow F(U) \leq \mathbb{R}^n$ is a diffeomorphism, then (U, F) is a chart.

Proof: The restriction $F : U \cap V \rightarrow F(U \cap V)$ is a diffeomorphism, and so is $\psi^{-1} : \psi(U \cap V) \rightarrow U \cap V$ for any chart (V, ψ) . Thus, the composition is also a diffeomorphism. A diffeomorphism between two subsets of $\mathbb{R}^n$ is smooth. The reverse direction is analogous.

4 Problems

1. $f : \mathbb{R} \rightarrow S^1$ given by $f(t) = (\cos t, \sin t)$ is C^∞ .
2. $\pi : S^1 \rightarrow \mathbb{R}$ given by $\pi(x, y) = x$ is C^∞ .
3. If $f_1 : M \rightarrow N_1, f_2 : M \rightarrow N_2$ are C^∞ , then so is $(f_1, f_2) : M \rightarrow N_1 \times N_2$.
4. If $f : M \rightarrow N$ is C^∞ and $U \leq M$ is open, then $f|_U$ is C^∞ .

1. I assume S^1 is given the standard atlas $\{(U_i, \phi_i)\}$, where $U_1 = f((0, \pi)), U_2 = f((\pi, 2\pi)), U_3 = f((-\pi/2, \pi/2)), U_4 = f((\pi/2, 3\pi/2)), \phi_1(x, y) = x, \phi_2(x, y) = x, \phi_3(x, y) = y, \phi_4(x, y) = y$. $\mathbb{R}$ has the atlas $\{(\mathbb{R}, 1_\mathbb{R})\}$. Let $V_i = f^{-1}(U_i)$, so V_i is open since f is continuous. We have $V_i \xrightarrow{1_\mathbb{R}^{-1}} V_i \xrightarrow{f} U_i \xrightarrow{\phi_i} (-1, 1)$. For $i = 1$, the map is $t \mapsto \cos t$ on an open subset of $\mathbb{R}$. Since C^∞ functions restrict to C^∞ functions on open subsets, the above composition is C^∞ . The other cases are similar. Since the U_i 's cover S^1 , the V_i 's cover $\mathbb{R}$.

2. $(-1, 1) \xrightarrow{\phi_i^{-1}} U_i \xrightarrow{\pi} \mathbb{R} \xrightarrow{1_\mathbb{R}} \mathbb{R}$ is smooth. For $i = 1$, the composition is inclusion $(-1, 1) \rightarrow \mathbb{R}$. For $i = 3$, the composition is $x \mapsto \sqrt{1 - x^2}$, which is smooth on $(-1, 1)$.

3. Let $p \in M$ and $(V_1, \psi_1), (V_2, \psi_2)$ be charts in N_1, N_2 containing $f_1 p, f_2 p$. Let (U, ϕ) be a chart in M containing p . We wish to show that the composition

$$\phi(U \cap f_1^{-1}(V_1) \cap f_2^{-1}(V_2)) \xrightarrow{\phi^{-1}} U \cap f_1^{-1}(V_1) \cap f_2^{-1}(V_2) \xrightarrow{(f_1, f_2)} V_1 \times V_2 \xrightarrow{\psi_1 \times \psi_2} \psi_1(V_1) \times \psi_2(V_2)$$

is C^∞ . This holds because $(\psi_1 \times \psi_2)(f_1, f_2)\phi^{-1} = (\psi_1 f_1 \phi^{-1}, \psi_2 f_2 \phi^{-1})$ is C^∞ .

4. Let (W, ϕ) be a chart in U , so (W, ϕ) is also a chart in M . Let (V, ψ) be a chart in N . The composition $\psi f \phi^{-1}$ is C^∞ .

Claim: $j : S^1 \rightarrow \mathbb{R}^2$ is C^∞ .

Proof: We have $\pi_1, \pi_2 : S^1 \rightarrow \mathbb{R}$ are C^∞ and $j = (\pi_1, \pi_2)$, so j is C^∞ .

Claim: If $(U, \phi) = (U, x^1, x^2, \dots, x^n)$ is a chart containing p , then $\{\frac{\partial f}{\partial x^i}|_p\}_{i \in [n]}$ is a basis of $T_p M$.

Proof: Suppose $\sum_{i \in [n]} a_i \frac{\partial}{\partial x^i}|_p = 0$. Applying the derivator to x^j , we have $a_j = 0$ since

$$\frac{\partial}{\partial x^i} x^j = \frac{\partial(r^j \circ \phi \circ \phi^{-1})}{\partial r^i}(\phi(p)) = \frac{\partial r^j}{\partial r^i}(\phi(p)) = \delta_{ij}.$$

Let $f : V \rightarrow \mathbb{R}$ be a representative of a germ with $\phi(V)$ star-shaped. Let $g = f \circ \phi^{-1}$ and write $g(y) = g(q) + \sum_{i \in [n]} (y^i - q^i) g_i(y)$, where $g_i(q) = \frac{\partial g}{\partial r^i}(q)$ and $q = \phi(p)$. Let X_p be a derivator at p . Set $y = \phi(x)$ for $x \in V$. Then

$$\begin{aligned} X_p(f(x)) &= X_p(g \circ \phi(x)) = \sum_{i \in [n]} X_p(y^i - q^i) g_i(\phi(p)) + (q^i - q^i) X_p(g_i \circ \phi(x)) \\ &= \sum_{i \in [n]} X_p(y^i - q^i) \frac{\partial g}{\partial r^i}(q) = \sum_{i \in [n]} a_i \frac{\partial(f \circ \phi^{-1})}{\partial r^i}(\phi(p)) = \sum_{i \in [n]} a_i \frac{\partial}{\partial x^i} \Big|_p f(x) \end{aligned}$$

for some $a_i \in \mathbb{R}$, so $X_p = \sum_{i \in [n]} \frac{\partial}{\partial x^i} \Big|_p$.

Claim: Let $(U, x^1, x^2, \dots, x^n), (V, y^1, y^2, \dots, y^n)$ be two charts on M . Then

$$\frac{\partial}{\partial x^j} = \sum_i \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^i}.$$

Claim: Let $c : (a, b) \rightarrow \mathbb{R}$ be a curve with target space $\mathbb{R}$. Then $c'(t) = \dot{c}(t) d/dx|_{c(t)}$.

Proof:

$$c'(t)(f) = c_*(d/dt)(f) = \frac{d}{dt}(f \circ c) = \dot{c}(t) \frac{d}{dx} \Big|_{c(t)} f.$$

Claim: Let $c : (a, b) \rightarrow M$ be a smooth curve and $(U, x^1, x^2, \dots, x^n)$ be a coordinate chart about $c(t)$. Write $c^i = x^i \circ c$. Then

$$c'(t) = \sum_{i \in [n]} \dot{c}^i(t) \frac{\partial}{\partial x^i} \Big|_{c(t)}.$$

Proof: $c'(t) = \sum_{i \in [n]} a_i(t) \frac{\partial}{\partial x^i} \Big|_{c(t)}$ for some $a_i(t)$. We compute

$$c'(t_0)(x^j) = c_{*, t_0}(d/dt|_{t_0})(x^j) = \frac{d}{dt} \Big|_{t_0} x^j \circ c(t) = \frac{d}{dt} \Big|_{t_0} c^j(t) = \dot{c}^j(t_0).$$

Thus $a_j(t_0) = \dot{c}^j(t_0)$.

Claim (existence of a curve with a given initial vector). For any $p \in M$ and $X_p \in T_p M$, there

exists $\epsilon > 0$ and smooth $c : (a, b) \rightarrow M$ such that $c(0) = p$ and $c'(0) = X_p$.

Proof: Let $(U, x^1, x^2, \dots, x^n)$ be a centered at p . Let $X_p = \sum_{i \in [n]} a^i \frac{\partial}{\partial x^i} \Big|_p$. Let $\alpha : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^n$ by $\alpha(t) = (a^1 t, a^2 t, \dots, a^n t)$, where $\epsilon > 0$ is such that $\alpha((-\epsilon, \epsilon)) \leq \phi(U)$. Let $c = \phi^{-1} \circ \alpha : (-\epsilon, \epsilon) \rightarrow M$, so $c(0) = \phi^{-1} \circ \alpha(0) = \phi^{-1}(0) = p$. Now

$$c'(0) = c_*(d/dt|_0) = (\phi^{-1})_* \alpha_*(d/dt|_0)$$

5 Problems

8.1. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the map $F(x, y) = (x, y, xy) = (u, v, w)$ and $p \in \mathbb{R}^2$ Compute $F_*(\partial/\partial x|_p)$ as a linear combination of $\partial/\partial u|_{F(p)}, \partial/\partial v|_{F(p)}, \partial/\partial w|_{F(p)}$.

Ans:

$$\begin{aligned} F_*(\partial/\partial x|_p)(u) &= \frac{\partial}{\partial x} \Big|_p u \circ F = \frac{\partial}{\partial x} \Big|_p x = 1 \\ F_*(\partial/\partial x|_p)(v) &= \frac{\partial}{\partial x} \Big|_p v \circ F = \frac{\partial}{\partial x} \Big|_p y = 0 \\ F_*(\partial/\partial x|_p)(w) &= \frac{\partial}{\partial x} \Big|_p w \circ F = \frac{\partial}{\partial x} \Big|_p xy = y(p) \end{aligned}$$

so

$$F_*(\partial/\partial x|_p) = \frac{\partial}{\partial u} \Big|_{F(p)} + y(p) \frac{\partial}{\partial w} \Big|_{F(p)}.$$

8.2. Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. For any $p \in \mathbb{R}^n$, there is a canonical isomorphism $T_p(\mathbb{R}^n) \cong \mathbb{R}^n$ given by

$$\sum_i a^i \frac{\partial}{\partial x^i} \Big|_p \mapsto \langle a^1, a^2, \dots, a^n \rangle.$$

Show that under this identification, $L_{*,p} : T_p(\mathbb{R}^n) \rightarrow T_p(\mathbb{R}^m)$ is L .

Ans: We wish to show that the following diagram commutes:

$$\begin{array}{ccc} T_p(\mathbb{R}^n) & \xrightarrow{L_{*,p}} & T_{L(p)}(\mathbb{R}^m) \\ \downarrow \varphi & & \downarrow \psi \\ \mathbb{R}^n & \xrightarrow{L} & \mathbb{R}^m \end{array}$$

where φ, ψ are the identifications above. Let $\mathbb{R}^n, \mathbb{R}^m$ have the bases and coordinate functions $\{e_j\}, \{f_i\}, \{x^j\}, \{y^i\}$. Assume $L(e_j) = \sum_{i \in [m]} l_{ij} f_i$. We compute

$$L_{*,p}(\partial/\partial x^j|_p)(y^i) = \frac{\partial}{\partial x^j} \Big|_p y^i \circ L = l_{ij}$$

so

$$L_{*,p}(\partial/\partial x^j|_p) = \sum_{i \in [m]} l_{ij} \frac{\partial}{\partial y^i} \Big|_{L(p)}.$$

It follows $\psi \circ L_{*,p}(\partial/\partial x^j|_p) = \langle l_{1j}, l_{2j}, \dots, l_{mj} \rangle$. For the other direction, we compute $L \circ \varphi(\partial/\partial x^j|_p) = L(e_j) = \sum_{i \in [m]} l_{ij} f_i = \langle l_{1j}, l_{2j}, \dots, l_{mj} \rangle$.

Claim: If M, N are manifolds, and π_1, π_2 are the projections of $M \times N$ onto M, N , then $(\pi_{1*}, \pi_{2*}) : T_{(p,q)}(M \times N) \rightarrow T_p(M) \times T_q(N)$ is an isomorphism.

Proof: Let $(U, x^1, x^2, \dots, x^m), (V, y^1, y^2, \dots, y^n)$ be charts in M, N containing p, q . For $1 \leq k \leq m$, let $z^k = x^k \circ \pi_1$ and for $m+1 \leq k \leq m+n$, let $z^k = y^{k-m} \circ \pi_2$. Then $(U \times V, z^1, z^2, \dots, z^{m+n})$ is a chart of $M \times N$ containing (p, q) . Suppose $Z_{(p,q)} \in T_{(p,q)}(M \times N)$ such that $\pi_{1*}(Z_{(p,q)}) = 0, \pi_{2*}(Z_{(p,q)}) = 0$. Write $Z_{(p,q)} = \sum_{k \in [m+n]} c^k \frac{\partial}{\partial z^k} \Big|_{(p,q)}$. Then for $1 \leq i \leq m$, we have

$$\pi_{1*}(Z_{(p,q)})(x^i) = Z_{(p,q)}(x^i \circ \pi_1) = Z_{(p,q)}(z^i) = c^i = 0.$$

A similar argument shows $c^i = 0$ for $m+1 \leq i \leq m+n$, so (π_{1*}, π_{2*}) is injective.

Now suppose $X_p = \sum_{i \in [m]} c^i \frac{\partial}{\partial x^i} \Big|_p \in T_p M$ and $Y_q = \sum_{j \in [n]} c^{j+m} \frac{\partial}{\partial y^j} \Big|_q \in T_q N$. Let $Z_{(p,q)} = \sum_{k \in [m+n]} c^k \frac{\partial}{\partial z^k} \Big|_{(p,q)}$. I claim that $(\pi_{1*}, \pi_{2*})(Z_{(p,q)}) = (X_p, Y_q)$. Indeed, for $i \in [m]$, we have

$$\begin{aligned} \pi_{1*}(Z_{(p,q)})(x^i) &= Z_{(p,q)}(x^i \circ \pi_1) = \sum_{k \in [m+n]} c^k \frac{\partial}{\partial z^k} \Big|_{(p,q)} (x^i \circ \pi_1) \\ &= \sum_{k \in [m+n]} c^k \frac{\partial}{\partial z^k} \Big|_{(p,q)} z^i = c^i \end{aligned}$$

so $\pi_{1*}(Z_{(p,q)}) = X_p$. The other case is similar.

8.8. Let G be a Lie group with identity e , multiplication $\mu : G \times G \rightarrow G$, and inverse $\iota : G \rightarrow G$.

(a). The differential at the identity $\mu_{*,(e,e)} : T_e G \times T_e G \rightarrow T_e G$ is addition: $\mu_{*,(e,e)}(X_e, Y_e) = X_e + Y_e$.

Proof: Let $X_e \in T_p G$. Pick $b : (-\epsilon, \epsilon) \rightarrow G$ such that $b(0) = e$ and $b'(0) = X_p$. Let $c : (-\epsilon, \epsilon) \rightarrow G \times G$ be $c = (b, 1_e)$, where $1_e : (-\epsilon, \epsilon) \rightarrow G$ is the constant path at e . I claim that $c'(0) = (X_e, 0)^*$. Suppose this is true. Then

$$\mu_{*,(e,e)}(X_e, 0) = \mu_{*,(e,e)} c_{*,0}(d/dt|_0) = b_{*,0}(d/dt|_0) = X_e.$$

Similarly, $\mu_{*,(e,e)}(0, Y_e) = Y_e$, so by linearity, $\mu_{*,(e,e)}(X_e, Y_e) = X_e + Y_e$.

Formally, we need to show that the following diagram commutes:

$$\begin{array}{ccc}
T_p N & \xrightarrow{(f_1, f_2)_*} & T_{(f_1 p, f_2 p)}(M_1 \times M_2) \\
& \searrow (f_{1*}, f_{2*}) & \downarrow (\pi_{1*}, \pi_{2*}) \\
& & T_{f_1 p} M_1 \times T_{f_2 p} M_2
\end{array}$$

Indeed, we can check each component: $\pi_{1*}(f_1, f_2)_* = f_{1*}$.

(b). Show that the differential of the inverse is negation: $\iota_{*,e}(X_e) = -X_e$.

Proof: Denote by $1_G : G \rightarrow G$ the identity. Observe that $\mu \circ (1_G, \iota) = e_e$ (sorry, different notation, e_e means constant function at e). By functoriality and part (a), $0 = \mu_{*,(e,e)} \circ (1_{G*,e}, \iota_{*,e}) = 1_{G*,e} + \iota_{*,e}$. Since $1_{G*,e} = 1_{T_e G}$, the result follows.

6 Submanifolds

9.1. Suppose $\frac{\partial f}{\partial x}(x, y) = 3x^2 - 6y = 0$ and $\frac{\partial f}{\partial y} = 2y - 6x = 0$. Then $2y = x^2$, so $x^2 - 6x = 0$, so $x = 0, 6$. The corresponding solutions are $(x, y, c) = (0, 0, 0), (6, 18, -108)$, so $f^{-1}c$ is regular submanifold for $c \neq 0, -108$. For $c = 0$, we solve

$$y = \frac{6x \pm \sqrt{36x^2 - 4x^3}}{2} = 3x \pm x\sqrt{9 - x}$$

For $c = -108$, we have $y^2 - 6xy + (x^3 + 108) = 0$, for which we solve

$$\begin{aligned}
y &= \frac{6x \pm \sqrt{36x^2 - 4(x^3 + 108)}}{2} = 3x \pm \sqrt{9x^2 - (x^3 + 108)} \\
&= 3x \pm (x - 6)\sqrt{-(x + 3)}
\end{aligned}$$

(we have to show that these arent topological manifolds).

9.3. Consider $F(x, y, z) = (x^3 + y^3 + z^3, xy - z)$. The Jacobian is

$$J(F) = \begin{bmatrix} 3x^2 & 3y^2 & 3z^2 \\ y & x & -1 \end{bmatrix}$$

which we want to be rank 2 for all x, y, z on the curve. Suppose that $J(f)$ is not rank 2, so the minors

$$\det \begin{bmatrix} 3x^2 & 3y^2 \\ y & x \end{bmatrix} = 3(x^3 - y^3), \det \begin{bmatrix} 3x^2 & 3z^2 \\ y & -1 \end{bmatrix} = -3x^2 - 3yz^2, \det \begin{bmatrix} 3y^2 & 3z^2 \\ x & -1 \end{bmatrix} = -3y^2 - 3xz^2$$

vanish (the last minor is guaranteed to be zero). Thus, $y = x$ and $z = x^2$, so $-3x^2 - 3x \cdot x^4 = 0 \rightarrow x = 0, -1$. We check that $(0, 0, 0), (-1, -1, 1)$ do not satisfy the equations, so the solution set

determines a regular submanifold by the level set theorem.

9.8. If S_i is a regular submanifold of the manifold M_i for $i = 1, 2$, then $S_1 \times S_2$ is a regular submanifold of $M_1 \times M_2$.

Proof: We assume the existence of compatible adaptable charts $\{(U, x^1, x^2, \dots, x^n)\}$ and $\{(V, y^1, y^2, \dots, y^m)\}$ of M_1, M_2 , so that $S_1 \cap U = \{p \in U : x^i(p) = 0 \text{ for } i > k\}$ and $S_2 \cap V = \{p \in V : y^j(p) = 0 \text{ for } j > l\}$ for some l, k (such that the U 's, V 's cover S_1, S_2 , resp).

I claim that $\{(U \times V, x^1 \circ \pi_1, x^2 \circ \pi_2, \dots, x^n \circ \pi_1, y^1 \circ \pi_2, y^2 \circ \pi_2, \dots, y^m \circ \pi_2)\}$ is a collection of compatible adapted charts of $S_1 \times S_2$. The $U \times V$'s covers $M_1 \times M_2$. Suppose that $x^i \circ \pi_1(p, q) = 0$ for $i > k$. Then $x^i(p) = 0$ for $i > k$, so $p \in S_1 \cap U$. Similarly, $q \in S_2 \cap V$. The converse is clear, so $(S_1 \times S_2) \cap (U \times V) = \{(p, q) \in U \times V : x^i \circ \pi_1(p, q) = 0, x^j \circ \pi_2(p, q) = 0 \text{ for } i > k, j > l\}$.

7 Lie Bracket

14.11. Compute $\left[-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}, \frac{\partial}{\partial x}\right]$. We check

$$\begin{aligned} \left[-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}, \frac{\partial}{\partial x}\right] x &= \left(-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}\right)(1) - \frac{\partial}{\partial x}(-y) = 0 \\ \left[-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}, \frac{\partial}{\partial x}\right] y &= \left(-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}\right)(0) - \frac{\partial}{\partial x}(x) = -1 \end{aligned}$$

so

$$\left[-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}, \frac{\partial}{\partial x}\right] = -\frac{\partial}{\partial y}$$

Ex: Let $X = x^2 d/dx$ on $\mathbb{R}$. Find the max integral curve of X starting at $x = 2$.

Proof: We want $c'(t) = c_*(d/dt|_0) = \dot{c}(t) \frac{d}{dx}|_{c(t)} = c(t)^2 \frac{d}{dx}|_{c(t)}$, so $\dot{c}(t) = c(t)^2$ for all t . If $c(t) \neq 0$ for all t , then $c(t) = -\frac{1}{t+a}$. Since $c(0) = 2$, we have $c(t) = \frac{2}{1-2t}$, defined on $(-\infty, 1/2)$.

8 Problems

Claim: Let $F : N \rightarrow M$ be a C^∞ diffeomorphism, X a C^∞ vector field on N , and g a C^∞ function on N . Then $F_*(gX) = (g \circ F^{-1})F_*(X)$.

Proof: Let f be C^∞ on M and $p \in N$. Then

$$\begin{aligned} (F_*(gX))_{Fp}(f) &= (gX)_p(f \circ F) = g(p)X_p(f \circ F) \\ ((g \circ F^{-1})F_*(X))_{Fp}(f) &= g(p)X_p(f \circ F) \end{aligned}$$

so the result holds.

Claim: Let $F : N \rightarrow M$ be a diffeomorphism of manifolds. Prove if X, Y are C^∞ vector fields on N , then $F_*[X, Y] = [F_*X, F_*Y]$.

Let f be C^∞ on M . First we show that $((F_*X)f) \circ F = X(f \circ F)$. Indeed,

$$(((F_*X)f) \circ F)(p) = (F_*X)_{Fp}(f) = X_p(f \circ F)$$

We have

$$\begin{aligned} [F_*X, F_*Y]_{Fp}(f) &= (F_*X)_{Fp}((F_*Y)f) - (F_*Y)_{Fp}((F_*X)f) \\ &= X_p(((F_*Y)f) \circ F) - Y_p(((F_*X)f) \circ F) \\ &= X_p(Y(f \circ F)) - Y_p(X(f \circ F)) \\ &= [X, Y]_p(f \circ F) = (F_*[X, Y])_{Fp}(f) \end{aligned}$$

showing the result.

9 (Tangent) Vector Bundles

Claim: Let $\pi : E \rightarrow M, \pi' : E' \rightarrow M$ and $\phi : E \rightarrow E'$. ϕ is fiber-preserving iff $\pi'\phi = \pi$.

Proof: Suppose ϕ is fiber-preserving, so $\phi(E_p) \subseteq E'_p$. Let $e \in E_p$. Then $\pi e = p$ and $\pi'\phi(e) = p = \pi e$, so $\pi'\phi = \pi$. Conversely, suppose $\pi'\phi = \pi$. Let $e \in E_p$. Then $\pi(e) = \pi'\phi(e) = p$, so $\phi(e) \in E'_p$.